EHM(X): 形式規則一覧

System F^{ξ} に集合効果 handler と静的述語 X を加える rank-1 表層

Figure 1. 構文・制約・効果の解釈

$$
\begin{aligned}
&e ::= x \mid n \mid \mathbf{true} \mid \mathbf{false} \mid \lambda x.e \mid e_1\, e_2 \mid \mathbf{let}\ x = e_1\ \mathbf{in}\ e_2 \mid \mathbf{perform}\ op\ e \mid \mathbf{handle}\ e\ \mathbf{with}\ h,\\
&h ::= \{\mathbf{return}\ r \mapsto e_r;\ \ op(x;k) \mapsto e_o\}, \qquad v ::= x \mid n \mid \mathbf{true} \mid \mathbf{false} \mid \lambda x.e,\\
&\tau ::= \alpha \mid \mathit{Int} \mid \mathit{Bool} \mid \tau \xrightarrow{\epsilon} \tau, \qquad \epsilon ::= \varnothing \mid \{op\} \mid \mu \mid \epsilon \cup \epsilon,\\
&\sigma ::= \forall \bar{\alpha}, \bar{\mu}.\ P \Rightarrow \tau,\\
&C ::= \top \mid C \wedge C \mid \tau = \tau \mid \epsilon \subseteq \epsilon \mid op \notin \epsilon \mid p_X \mid \forall \alpha.C \mid \forall \mu.C.
\end{aligned}
$$

$$
\epsilon \equiv \epsilon' \quad \overset{\text{def}}{\Longleftrightarrow} \quad (\epsilon \subseteq \epsilon') \wedge (\epsilon' \subseteq \epsilon).
$$

$$
\mathrm{NF}(\epsilon) = (L_\epsilon, V_\epsilon), \qquad I_\rho(\epsilon) = L_\epsilon \cup \bigcup_{\mu \in V_\epsilon} \rho(\mu), \qquad I_\rho(\{\mathit{Read}\} \cup \mu) = \{\mathit{Read}\} \cup \rho(\mu).
$$

$$
\boxed{\mathrm{Split}(op, \epsilon, \delta) \overset{\text{def}}{=} (\delta \subseteq \epsilon) \wedge (\epsilon \subseteq \{op\} \cup \delta) \wedge (op \notin \delta)}
$$

$$
\rho \models \mathrm{Split}(op, \epsilon, \delta) \quad \Longleftrightarrow \quad I_\rho(\delta) = I_\rho(\epsilon) \setminus \{op\}.
$$

Figure 2. 値呼びの簡約規則

$$
(\lambda x.e)\, v \longrightarrow e[x \mapsto v] \qquad \mathbf{handle}\ v\ \mathbf{with}\ h \longrightarrow e_r[r \mapsto v].
$$

$$
\begin{aligned}
&\mathbf{handle}\ E[\mathbf{perform}\ op\ v]\ \mathbf{with}\ h \longrightarrow e_o[x \mapsto v][k \mapsto k_E],\\
&op \notin \mathrm{handled}(E), \qquad k_E = \lambda y.\mathbf{handle}\ E[y]\ \mathbf{with}\ h.
\end{aligned}
$$

宣言的型付け　$C; \Gamma \vdash e : \tau \mid \epsilon$

Figure 3. 基本規則

$$\frac{\Gamma(x) = \forall \bar{\beta}.P \Rightarrow \tau \quad C \models P[\bar{\theta}/\bar{\beta}]}{C; \Gamma \vdash x : \tau[\bar{\theta}/\bar{\beta}] \mid \varnothing} \quad (\textsc{Var})$$

$$\frac{C; \Gamma, x : a \vdash e : b \mid \eta}{C; \Gamma \vdash \lambda x.e : a \xrightarrow{\eta} b \mid \varnothing} \quad (\textsc{Abs})$$

$$\frac{C; \Gamma \vdash e_1 : a \xrightarrow{\eta} b \mid \epsilon_1 \quad C; \Gamma \vdash e_2 : a \mid \epsilon_2}{C; \Gamma \vdash e_1\, e_2 : b \mid \epsilon_1 \cup \epsilon_2 \cup \eta} \quad (\textsc{App})$$

$$\frac{\Sigma(op) = \forall \bar{\beta}.\tau_a \to \tau_b \qquad C; \Gamma \vdash e : \tau_a[\bar{\theta}/\bar{\beta}] \mid \eta}{C; \Gamma \vdash \textbf{perform}\ op\ e : \tau_b[\bar{\theta}/\bar{\beta}] \mid \eta \cup \{op\}} \quad (\textsc{Perform})$$

$$\frac{n \in \mathbb{Z}}{C; \Gamma \vdash n : \mathit{Int} \mid \varnothing} \quad (\textsc{Int}) \qquad \frac{b \in \{\textbf{true}, \textbf{false}\}}{C; \Gamma \vdash b : \mathit{Bool} \mid \varnothing} \quad (\textsc{Bool})$$

Figure 4. 値制限付き let

$$\frac{C_1; \Gamma \vdash v : \tau_1 \mid \varnothing \quad A = \mathrm{FV}(C_1, \tau_1) \setminus \mathrm{FV}(\Gamma) \quad \sigma = \forall A.C_1 \Rightarrow \tau_1 \quad C_2; \Gamma, x : \sigma \vdash e_2 : \tau_2 \mid \epsilon_2}{C_2; \Gamma \vdash \textbf{let}\ x = v\ \textbf{in}\ e_2 : \tau_2 \mid \epsilon_2} \quad (\textsc{LetVal})$$

$$\frac{\neg\, \mathrm{Value}(e_1) \quad C; \Gamma \vdash e_1 : \tau_1 \mid \epsilon_1 \quad C; \Gamma, x : \tau_1 \vdash e_2 : \tau_2 \mid \epsilon_2}{C; \Gamma \vdash \textbf{let}\ x = e_1\ \textbf{in}\ e_2 : \tau_2 \mid \epsilon_1 \cup \epsilon_2} \quad (\textsc{LetMono})$$

Figure 5. Deep handler

$$\frac{\begin{array}{c}\Sigma(op) = \forall \overline{\beta :: \kappa}.\tau_a \to \tau_b \qquad C; \Gamma \vdash e : \tau_{in} \mid \epsilon_{in} \\ C; \Gamma, r : \tau_{in} \vdash e_r : \tau_{out} \mid \epsilon_r \\ C; \Gamma, x : \tau_a, k : \tau_b \xrightarrow{\chi} \tau_{out} \vdash e_o : \tau_{out} \mid \epsilon_o \\ C \models \mathrm{Split}(op, \epsilon_{in}, \delta) \quad C \models \delta \subseteq \chi \quad C \models \epsilon_r \subseteq \chi \quad C \models \epsilon_o \subseteq \chi\end{array}}{C; \Gamma \vdash \textbf{handle}\ e\ \textbf{with}\ \{\textbf{return}\ r \mapsto e_r;\ op(x; k) \mapsto e_o\} : \tau_{out} \mid \chi} \quad (\textsc{Handle})$$

$\bar{\beta}$ rigid in the operation clause, $\qquad \bar{\beta} \cap \mathrm{FV}(\Gamma, \tau_{in}, \tau_{out}, \epsilon_{in}, \epsilon_r, \chi, \delta) = \varnothing.$

制約生成　$\mathrm{Gen}_\Gamma(e) = (\tau, \epsilon, C)$

Figure 6. 構文に沿った生成方程式

$$\mathrm{Gen}_\Gamma(n) = (\mathit{Int}, \varnothing, \top), \qquad \mathrm{Gen}_\Gamma(b) = (\mathit{Bool}, \varnothing, \top) \quad (b \in \{\mathbf{true}, \mathbf{false}\}).$$

$$\Gamma(x) = \sigma = \forall \bar{\beta}.P \Rightarrow \tau_x, \quad \theta \text{ は } \bar{\beta} \text{ の許容される改名} \Longrightarrow \mathrm{Gen}_\Gamma(x) = (\theta\tau_x, \varnothing, \theta P),$$

$$\mathrm{Gen}_{\Gamma, x:a}(e) = (b, \eta, C) \Longrightarrow \mathrm{Gen}_\Gamma(\lambda x.e) = (a \xrightarrow{\eta} b, \varnothing, C).$$

$$\mathrm{Gen}_\Gamma(e_1) = (\tau_1, \epsilon_1, C_1), \quad \mathrm{Gen}_\Gamma(e_2) = (\tau_2, \epsilon_2, C_2) \Longrightarrow \mathrm{Gen}_\Gamma(e_1\, e_2)$$
$$= (b, \epsilon_1 \cup \epsilon_2 \cup \eta, C_1 \wedge C_2 \wedge (\tau_1 = \tau_2 \xrightarrow{\eta} b)).$$

$$\Sigma(\mathit{op}) = \forall \bar{\beta}.\tau_a \to \tau_b, \quad \mathrm{Gen}_\Gamma(e) = (\tau, \eta, C) \Longrightarrow \mathrm{Gen}_\Gamma(\mathbf{perform}\ \mathit{op}\ e)$$
$$= (\theta\tau_b, \{\mathit{op}\} \cup \eta, C \wedge (\tau = \theta\tau_a)).$$

a, b, η, θ は使用時に新たに選ぶ。

Figure 7. let の生成と局所的な一般化

$$\mathrm{Gen}(\Gamma, v) = (\tau_1, \varnothing, C_1), \qquad A = \mathrm{FV}(\tau_1, C_1) \setminus \mathrm{FV}(\Gamma),$$
$$\sigma = \forall A.C_1 \Rightarrow \tau_1, \qquad \mathrm{Gen}_{\Gamma, x:\sigma}(e_2) = (\tau_2, \epsilon_2, C_2).$$

$$\mathrm{Gen}_\Gamma(\mathbf{let}\ x = v\ \mathbf{in}\ e_2) = (\tau_2, \epsilon_2, C_2),$$

$$\mathrm{Gen}_\Gamma(e_1) = (\tau_1, \epsilon_1, C_1), \quad \mathrm{Gen}_{\Gamma, x:\tau_1}(e_2) = (\tau_2, \epsilon_2, C_2) \Longrightarrow \mathrm{Gen}_\Gamma(\mathbf{let}\ x = e_1\ \mathbf{in}\ e_2)$$
$$= (\tau_2, \epsilon_1 \cup \epsilon_2, C_1 \wedge C_2), \quad \neg\, \mathrm{Value}(e_1).$$

Handler の生成・解決・健全性

Figure 8. Handler の制約生成

$$\Sigma(op) = \forall\overline{\beta :: \kappa}.\tau_a \to \tau_b, \qquad a, \delta, \chi \text{ fresh},$$
$$\mathrm{Gen}_\Gamma(e) = (\tau_{in}, \epsilon_{in}, C_{in}),$$
$$\mathrm{Gen}_{\Gamma, r:\tau_{in}}(e_r) = (\tau_r, \epsilon_r, C_r),$$
$$\mathrm{Gen}_{\Gamma, x:\tau_a, k:\tau_b \xrightarrow{\chi} a}(e_o) = (\tau_o, \epsilon_o, C_o).$$

$$\begin{aligned}\mathrm{Gen}_\Gamma(\textbf{handle } e \textbf{ with } h) = (a, \chi, C),&\\ C = C_{in} \wedge C_r \wedge (\tau_r = a)&\\ \wedge \forall\bar{\beta}.(C_o \wedge \tau_o = a \wedge \epsilon_o \subseteq \chi)&\\ \wedge \mathrm{Split}(op, \epsilon_{in}, \delta) \wedge \delta \subseteq \chi \wedge \epsilon_r \subseteq \chi.&\end{aligned}$$

$$\bar{\beta} \text{ rigid}; \qquad \bar{\beta} \cap \mathrm{FV}(\Gamma, \tau_{in}, a, \delta, \chi, \epsilon_{in}, \epsilon_r) = \varnothing.$$

Figure 9. 解決器の保存条件

$$(a_1 \xrightarrow{\eta_1} b_1) = (a_2 \xrightarrow{\eta_2} b_2) \rightsquigarrow (a_1 = a_2) \wedge (b_1 = b_2) \wedge (\eta_1 \equiv \eta_2),$$
$$\mathrm{simplify}(C) = \mathrm{some}(R) \Longrightarrow \ \forall\rho.\ \rho \models R \Rightarrow \rho \models C,$$
$$\mathrm{resolve}_X(A, C) = R_X \Longrightarrow \ \forall\rho.\ \rho \models A \wedge R_X \Rightarrow \rho \models C.$$

型変数の単一化・principal type・操作的安全性はこの証明の対象外。

Figure 10. *ping* **の制約簡約**

$$\Sigma(ping) = \forall\beta.\beta \to \beta, \qquad f : \forall\alpha.\ Eq\,\alpha \Rightarrow \alpha \xrightarrow{\varnothing} \alpha,$$
$$g = \lambda x.\textbf{perform } ping\,(f\,x), \qquad g : \forall\alpha.\ Eq\,\alpha \Rightarrow \alpha \xrightarrow{\{ping\}} \alpha,$$
$$e = \textbf{handle } (g\,y) \textbf{ with } \{\textbf{return } r \mapsto r;\ ping(x; k) \mapsto k\,x\},$$
$$\mathrm{Split}(ping, \{ping\}, \delta) \Longrightarrow \delta = \varnothing, \qquad \epsilon_r = \varnothing, \qquad \epsilon_o = \chi,$$
$$C_e = Eq\,\alpha \wedge (\varnothing \subseteq \chi) \wedge (\chi \subseteq \chi), \qquad \chi = \varnothing \text{ is one solution.}$$

1　主要な補題・命題・定理

$$R \models C \overset{\text{def}}{\iff} \forall \rho.\ \rho \models R \Rightarrow \rho \models C, \qquad \mathrm{Assume}(A) = \bigwedge_{p \in A} p.$$

補題 1.1 (操作の厳密な除去)**.** 任意の評価 ρ について,

$$\rho \models \mathrm{Split}(op, \epsilon, \delta) \quad \Longleftrightarrow \quad I_\rho(\delta) = I_\rho(\epsilon) \setminus \{op\}.$$

補題 1.2 (剛性変数の全称閉包)**.** 型変数・効果変数の有限集合 B_T, B_E について,

$$\mathrm{Close}_{B_T,B_E}(D) \overset{\text{def}}{=} \forall B_T.\forall B_E.D, \qquad \mathrm{Close}_{B_T,B_E}(D) \models D.$$

命題 1.3 (多相操作引数の固定結果型への漏出)**.** $i \neq j$ ならば,

$$\neg\exists \rho.\ \rho \models \forall \alpha_i.(\alpha_i = \alpha_j).$$

補題 1.4 (値の生成効果)**.**

$$\mathrm{Value}(v) \ \wedge\ \mathrm{Gen}_\Gamma(v) = (\tau, \epsilon, C) \quad \Longrightarrow \quad \epsilon = \varnothing.$$

命題 1.5 (宣言的型付けに対する生成規則の健全性)**.**

$$\mathrm{Gen}_\Gamma(e) = (\tau, \epsilon, C) \quad \Longrightarrow \quad C; \Gamma \vdash_{\mathrm{EHM}(X)} e : \tau \mid \epsilon.$$

命題 1.6 (実行可能な生成器の規則への適合)**.**

$$\mathrm{generate}(\Sigma, X, \Gamma, e, s) = \mathrm{some}(r) \quad \Longrightarrow \quad \mathrm{Gen}_\Gamma(e) = (r.\tau, r.\epsilon, r.C).$$

補題 1.7 (簡約器の健全性)**.**

$$\mathrm{simplify}(C) = \mathrm{some}(R) \quad \Longrightarrow \quad R \models C.$$

補題 1.8 (追加述語の解決の健全性)**.**

$$\mathrm{Assume}(A) \wedge \mathrm{resolve}_X(A, C) \models C.$$

定理 1.9 (EHM(X) 推論の健全性)**.**

$$\mathrm{inferQualified}(\Sigma, X, A, \Gamma, e, s) = \mathrm{some}(r) \quad \Longrightarrow \quad r.C; \Gamma \vdash_{\mathrm{EHM}(X)} e : r.\tau \mid r.\epsilon.$$

ここで $r.C$ は $\mathrm{Assume}(A)$ と未解決の制約を含む.

$ping(x; k) \mapsto x$ は上の命題により充足不能となる．$ping(x; k) \mapsto k\,x$ では $\chi = \varnothing$ は一解だが，この推論器は最小効果を選ばず χ と包含制約を残す.

対応する Lean 定理：`Constraint.split_iff`, `Constraint.close_current`, `Constraint.rigid_var_eq_unsat`.
`Generate.value_effect`, `Generate.sound`, `generate_sound`.
`simplify_sound`, `resolveX_sound`, `inferQualified_sound`.

2　導出木：$ping(x;k) \mapsto x$ の制約

$$\Sigma(ping) = \forall\alpha.\alpha \to \alpha, \qquad \sigma_{id} = \forall\gamma.\gamma \xrightarrow{\varnothing} \gamma, \qquad T = \theta \xrightarrow{\varnothing} \theta,$$
$$B = \textbf{let } id = \lambda z.z \textbf{ in perform } ping\ id,$$
$$H = \textbf{handle } B \textbf{ with } \{\textbf{return } r \mapsto r;\ ping(x;k) \mapsto x\}.$$

本体の導出． id は σ_{id} から T に具体化し，**perform** でも操作型引数を $\alpha := T$ とする．

$$\dfrac{\dfrac{\dfrac{}{z:\theta \vdash z:\theta \mid \varnothing}\,(\text{Abs})}{\vdash \lambda z.z : T \mid \varnothing} \qquad \dfrac{}{id:\sigma_{id} \vdash id : T \mid \varnothing}\,(\text{Perform } \alpha := T)}{\vdash B : T \mid \{ping\}}\,(\text{LetVal})$$

宣言的な Handle 規則の適用． 本体の前提は上で得た．規則全体には return 節と操作節の前提も要る．

$$\dfrac{C;\varnothing \vdash B : T \mid \{ping\} \qquad C; r:T \vdash r : a \mid \varnothing \qquad C; x:\beta,\ k:\beta \xrightarrow{\chi} a \vdash x : a \mid \varnothing}{C;\varnothing \vdash H : a \mid \chi}\,(\text{Handle})$$

第 2 前提は $C \models T = a$，第 3 前提は $C \models \beta = a$ を要求する．さらに Split と効果包含が必要である．従って B の型だけでは結論を得られない．

Handler の制約生成． 操作節では別の剛性変数 β を導入する．a は handler の結果型で，$\beta \notin \mathrm{FV}(a)$ である．

$$\dfrac{\mathrm{Gen}_{\varnothing}(B) = (T, \{ping\}, C_B) \qquad \mathrm{Gen}_{r:T}(r) = (T, \varnothing, \top) \qquad \mathrm{Gen}_{x:\beta,\ k:\beta \xrightarrow{\chi} a}(x) = (\beta, \varnothing, \top)}{\mathrm{Gen}_{\varnothing}(H) = (a, \chi, C_H)}\,(\text{Handle})$$

$$C_H = C_B \wedge (T = a) \wedge \underbrace{\forall\beta.(\beta = a \wedge \varnothing \subseteq \chi)}_{\text{操作節の結果型 } \tau_o = a}$$
$$\wedge\, \mathrm{Split}(ping, \{ping\}, \delta) \wedge \delta \subseteq \chi \wedge \varnothing \subseteq \chi.$$

$$\mathrm{Split}(ping, \{ping\}, \delta) \Longrightarrow \delta = \varnothing, \qquad \boxed{\forall\beta.\beta = a \quad \text{は充足不能}}.$$

$\beta = Int$ と $\beta = Bool$ を選ぶと，同じ固定型 a が Int と $Bool$ の両方に等しくなって矛盾する．$C = \top$ では Handle 規則の第 3 前提を証明できない．$C = C_H$ とすれば形式上は規則を適用できるが，C_H を満たす評価は存在しない．実際の出力では $\theta = \alpha_{10}$, $T = \alpha_6$, $\beta = \alpha_0$, $a = \alpha_{11}$, $\delta = \mu_7$, $\chi = \mu_8$ に対応する．C_B 内の $(\alpha_{10} \xrightarrow{\varnothing} \alpha_{10}) = \alpha_6$ を解いた形で本体の導出木を表示した．

$ping(x;k) \mapsto k\,x$ なら，$x:\beta$ と $k:\beta \xrightarrow{\chi} a$ から節の結果型は a となり，$\forall\beta.\beta = a$ は生成されない．